

Statistics Ch. 6

Statistics

Random Variables

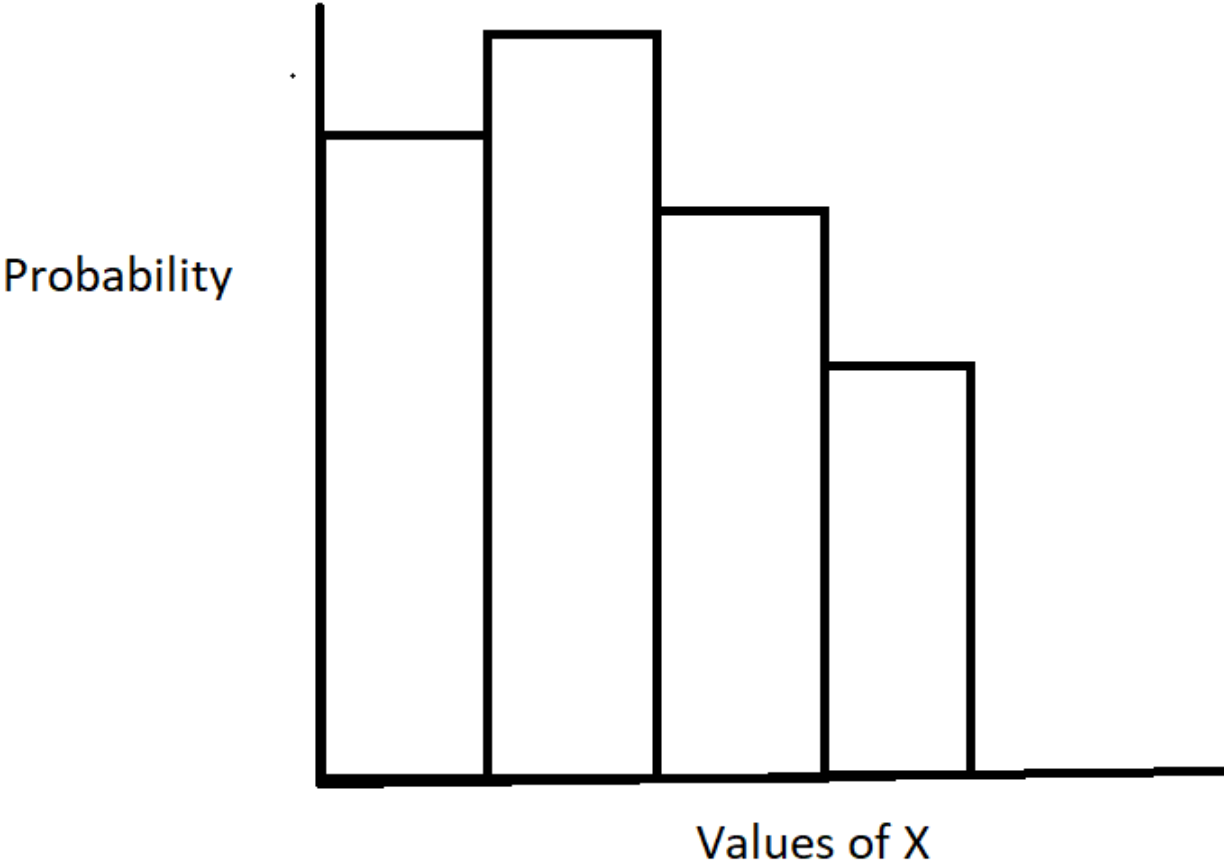
Probability Model - possible outcomes of a chance process and their probabilities

Random Variable - numerical value that describes the outcome of some chance process

- Capital letter X, Y, W
Probability Distribution of a R.V - possible values and their probabilities
Two types of RV
- Discrete and Continuous

Discrete

- Takes on a fixed number of values with gaps between
 - x _ _ _ _
 - P(x) _ _ _ _
- Ex. How many children: 1, 2, 3. notice how there are no decimals, gaps in between the whole nums
 - Display with Histogram



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- Mean or Expected Value = $\sum x_i p_i$ (weighted average)
to find expected value on your calculator
[Stat] -> Calc -> 1 Var. Stats
 L_1 = outcomes
 L_2 = freq. (prob)
(find L_1 and L_2 in [Stat] -> edit)

Continuous

- takes on all possible values in an interval
- display with density curve (one type is normal)
- individual outcomes has a probability of 0
- only intervals have a positive probability
- probability is area under the curve

Standard Deviation

std. dev = $\sqrt{\sum (x_1 - \mu)^2 p_i}$

variance = $\sigma_x^2 = \sum (x_1 - \mu)^2 p_i$

$Z = \frac{x - \mu}{\sigma}$ Table A

normcdf()



Transforming Probability Distributions

RV

adding/subtracting constant c

- Shape: stays the same
- Center: add/sub by c
- Spread: stays the same

multiply/dividing constant c

- Shape: stays the same
- Center: mult/div by c
- Spread: mult/div by c

Add RV X and Y

$$\mu_{x+y} = \mu_x + \mu_y$$
$$\sigma_{x+y} = \sqrt{\sigma_x^2 + \sigma_y^2}$$

Sub RV X and Y

$$\mu_{x-y} = \mu_x - \mu_y$$
$$\sigma_{x-y} = \sqrt{\sigma_x^2 + \sigma_y^2}$$

Prob Dist --> find mean and SD

Binomial RV.

2 types of discrete RV.

1. Binomial
2. Geometric

Binomial RV.

- Conditions --> Binomial Setting
- **BINS**:
 - **B**: Binary - two possible outcomes
 - **I**: Independent - knowing success of one outcome does not affect outcome of other trial
 - **N**: Number - fixed number of trials
 - **S**: Success - same probability of success on each trial

Binomial Probability Formula

X has a binomial distribution, with

n = trials

p = probability of success

$$P(X = K) = \binom{n}{k} p^k (1 - p)^{n-k}$$

where p^k is prob of success, $(1 - p)^{\{n-k\}}$ is prob of failure, and

$$\binom{n}{k} = \frac{n!}{k!(1-k)!}$$

you can use calculator through [2nd][Vars] dist --> binomial

binomialpdf(n,p,k) = P(X=K)

binomialcdf(n,p,k) = P(X≤K)

$$\text{Mean} = \mu_x = np$$

After many many trials the average number of success *context* is μ of n

$$\text{SD} = \sigma_x = \sqrt{np(1 - p)}$$

The number of success typically varies by σ_x from the mean of n

Normal Approximations to Binomial Dist

Conditions:

1. 10% Rule
 - When taking random sample (without replacement) of size n from population size N , we can use a binomial distribution if $n \leq \frac{1}{10} N$
2. Large Counts Condition
 - Use a normal distribution to model a binomial if $np \geq 10$ (expected # of success) and $n(1 - p) \geq 10$ (expected # of failures)

Geometric

BITS

- **B**: Binary
- **I**: Independent
- **T**: Trials until a success
- **S**: Same probability

$$P(X = k) = (1 - p)^{\{k-1\}} p$$

- $(1 - p)$: prob of fail
- $k - 1$: num of fail
- p : prob of success

$$P(X = K) = \text{geompdf}(p,k)$$

$$P(X \leq K) = \text{geompdf}(p,k)$$

Describe:

- Shape --> right skew
- center, mean = $\mu = \frac{1}{p}$
- SD = $\sigma_x = \frac{\sqrt{1-p}}{p}$